



Intoduction To Programming Using C++ Level 2 Ranged Loop



Introduction To Programming Using C++

Level 2

Ranged Loop

Definition

- ◆ A Ranged `for` Loop is a loop used to iterate through the elements of a collection, such as an Array or Vector.
- ◆ It allows us to access each element directly without using an Index.

Explanation

- A Ranged `for` Loop uses a variable that represents the current element.
- The loop automatically moves from one element to the next.
- It can be used with Arrays, Vectors, and other suitable collections.
- It is useful when we want to process every element without manually managing the Index.
- We do not need to know or manually calculate the number of elements when using a suitable collection with ranged iteration.

Procedures

1. Create an Array, Vector, or another suitable collection.
2. Write the ``for`` keyword.
3. Declare a variable that will represent the current element.
4. Use the ``:`` operator.
5. Specify the collection that we want to iterate through.
6. Write the code that should execute for each element.
7. The loop automatically moves to the next element.
8. The loop stops after all elements have been processed.

💡 Example

◆ Suppose we have:

```
int arr[] = {1, 2, 3, 4};  
for (int n : arr)  
{  
    cout << n << endl;  
}
```

◆ In each iteration, `n` contains the current element of the Array.

◆ The loop will process:

◆ `n = 1`

◆ `n = 2`

◆ `n = 3`

◆ `n = 4`

➡ Result

The program prints:

1

2

3

4

Note

- ◆ A Ranged `for` Loop is especially useful when we want to access every element without using an Index such as `arr[i]`.
- ◆ It can be used with different collections, including Arrays and Vectors.

⊘ Warning

◆ If you need the Index of each element, a traditional `for` Loop may be more appropriate, or you can use another technique to track the Index.



Comparison

▨ Ranged `for` Loop

- ✓ Accesses elements directly.
- ✓ Does not require manually managing an Index.
- ✓ Makes simple iteration shorter and easier to read.

▤ Traditional `for` Loop

- ✓ Gives direct control over the Index.
- ✓ Useful when we need the position of each element.
- ✓ Useful when we need more control over the iteration process.

✦✦ Summary

- ✓ A Ranged `for` Loop iterates directly through the elements of a collection.
- ✓ It can be used with Arrays, Vectors, and other suitable collections.
- ✓ It is useful when we want to process every element without manually using an Index.

Thank you 😊